INCH-POUND

MIL-P-44412(GL)
11 December 1990

MILITARY SPECIFICATION

PACK, PARACHUTIST'S, STINGER MISSILE

This specification is approved for use by the Natick Research, Development, and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers one type and size of a parachutist's reinforced jump pack to carry the Stinger Missile. The jump pack is fabricated from nylon duck and nylon webbing.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herin. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solication (see 6.2).

SPECIFICATIONS

FEDERAL

A-A-50083 - Bag, Plastic, Folded Garment
PPP-B-601 - Boxes, Wood Cleated-Plywood
PPP-B-636 - Boxes, Shipping, Fiberboard

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5017 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1670

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads
MIL-STD-731 - Quality of Wood Members for Containers and Pallets

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

U.S. ARMY NATICK RESEARCH, DEVELOPMENT, AND ENGINEERING CENTER

11-1-3890 - Jump Pack, Stinger Missile Main Assembly

(Copies of drawings are available from the U.S. Army Natick Research, Development, and Engineering Center, ATTN: STRNC-UX, Natick, MA 01760-5017.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the soliciation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard practice for Commercial Packing

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

(Non-Government standards and other publications are normally available from organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited hererin, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.2 Materials and components. Materials and components shall conform to the requirements on the applicable drawings, to subsidiary specifications and standards applicable thereto and as specified herein. It is encouraged that recycled material be used when practical as long as it meets the requirements of this specification.

3.3 Construction. The construction shall conform in all respects to the drawings listed in Section 2 and as specified herein.

3.3.1 Type 301 and 308 stitching. Ends of all type 301 stitching shall be backstitched or overstitched not less than 1/2 inch in accordance with the required stitch pattern. Ends of type 308 stitching shall be secured by crowding a minimum of three stitches (accomplished by not allowing the materials to move) at the beginning and end of the stitching. Thread tension shall be maintained so that there will be no loose stitching resulting in the puckering of the material sewn. The lock shall be embedded in the materials sewn.

3.3.1.1 Repairs of type 301 stitching. Repairs of type 301 stitching shall be as follows:

a. When thread breaks or bobbin run-outs occur during the stitching, the stitching shall be repaired by restarting the stitching a minimum of 1 inch (1/2 inch for box-x and W-W stitching) from the end of the stitching. 1/

b. Thread breaks or two or more consecutive skipped or run-off stitches noted during inspection of the item (in-process or end item) shall be repaired by overstitching. The stitching shall start a minimum of 1 inch from the defective area, (1/2 inch for box-x and W-W stitching), continue over the defective area, and continue a minimum of 1 inch (1/2 inch for box-x and W-W stitching) beyond the defective area onto the existing stitching. Loose or excessively tight stitching shall be repaired by removing the defective stitching, without damaging the materials, and restitching in the required manner. 1/

1/ When making the above repairs, the ends of the stitching are not required to be backstitched.

3.3.1.2 Repair of 308 stitching. Repairs of type 308 stitching shall be in accordance with 3.3.1.1.a and 3.3.1.1.b, except that the stitching shall overstitch the defective area a minimum of three stitches onto the existing stitches.

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3.3.1.3 Automatic stitching. Automatic stitching machines may be used to perform any of the stitch patterns provided the requirements for the stitch pattern, stitches per inch, and size and type of thread are met; and at least three or more tying, overlapping, or back stitches are used to secure the ends of the stitching.

3.4 Thread ends. All thread ends shall be trimmed to a maximum length of 1/4 inch.

3.5 Setting of grommets. Holes shall be prepunched to receive the grommets. Holes prepunched to receive the grommets shall be smaller than the outside diameter of the grommet barrel so that the barrel must be forced through the hole. The grommet shall be securely clinched without cutting the adjacent material.

3.6 Repairs. Repairs such as mends, darns, patches, or splices are not permitted on the pack.

3.7 Replacement of defective components. During the spreading, cutting and manufacturing process, components having material defects or damages that are classified as defects in 4.4.3, shall be removed from production and replaced with nondefective and properly matched components.

3.8 Workmanship. The pack shall conform to the quality of product established by this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

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e acceptable quality levels shall not exceed the applicab

4.1.2 Repsonsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for ensuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point, or at all points in the manufacturing process necessary to ensure compliance with all dimensional requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 3.1 and 6.2), it shall be examined for the defects specified in 4.4.3 and 4.4.4. Any nonconformance shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase documents.

4.4.2 In-process inspection. Inspection of subassemblies shall be made to ascertain that construction details which cannot be examined in the finished product are in accordance with specified requirements. The Government reserves the right to exclude from consideration for acceptance, any material or service for which in-process inspection has indicated nonconformance.

4.4.3 End item visual examination. The end items shall be examined for the defects listed in table I. The lot size shall be expressed in units of packs. The sample unit shall be one pack. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 1.0 for major defects and 6.5 for total (major and minor combined) defects.

TABLE I. End item visual defects.

Examine	Defect	Classification Major	Minor
Fabric	Any hole (except drill holes), cut, or tear, weak places or broken yarns	101	
	Exposed drill holes		201
Felt	Any hole, cut, or tear		202

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TABLE I. End item visual defects. (cont'd)

Examine	Defect	Classification Major	Minor
Webbing and tape	Not firmly and tightly woven; edge frayed or scalloped	102	
	Exposed end of tie down tapes not dipped in wax as required		203
	Exposed end of nylon webbing or tape not seared smooth as required		204
Fastener tape, hook and pile	Any hole, cut, or tear. Hooks missing or flattened, foreign matter or dirt impairing proper function	103	
Hardware	Broken or bent; corroded areas; burrs or sharp edges	104	
	Any required component not specified type or size	105	
	Any required component not functioning, improperly set or reversed	106	
Brass grommets	Clinched excessively tight, cutting adjacent material	107	
	Insecurely clinched to a degree that grommet may be detached from material	108	
	Washer installed on incorrect side of material		205
	Barrel split		206
	Loosely clinched, permitting grommet to rotate freely within the hole		207
	Washer omitted	109	
Open seams and run-offs	1/2 inch or less		208
	More than 1/2 inch	110	

NOTE: A seam shall be classified as open when one or more stitches joining a seam are broken or when two or more skipped stitches or run-offs occur. On a double stitching seams, a seam shall be considered open when either one or both sides of the seam are open.

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Cut Length Keeper Shape

TABLE I. End item visual defects. (cont'd)

Examine	Defect	Classification Major	Classification Minor
Binding	Improperly or loosely applied showing raw edge of fabric	111	
	End of binding not finished as specified or overlapped less than 1 inch		209
Seam and stitch type	Wrong seam or stitch type	112	
Stitch tension	Loose, resulting in a loosely exposed top or bobbin thread; tight, resulting in excessive tightness of fabric on seams	113	
Stitches per inch	One stitch less than minimum specified		210
	Two or more stitches less than minimum specified	114	
	One or two stitches in excess of maximum specified		211
	More than two stitches in excess of maximum specified	115	
	NOTE: Variation in the number of stitches per inch, caused by the operator speeding up the machine and pulling the fabric in order to sew over heavy places, heavy seams or in turning corners, shall be classified as follows: a. Within the major defect classification - minor defect b. Within the minor defect classification - no defect		
Stitching margins	Larger than specified by more than 3/16 inch for more than 2 inches		212
Thread breaks	Not overstitched as specified		213
	NOTE: Thread breaks not overstitched shall be classified as open seams		
Stitching ends	Ends of stitching backstitched less than 1/2 inch (except where ends are turned under in a hem or held down by other stitching)		214

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Stitch tension Loose, resulting in loosely ex-

TABLE I. End item visual defects. (cont'd)

Examine	Defect	Classification Major	Minor
Stitching ends (cont'd)	Ends of stitching not backstitched	116	
Stitching	Any required row of stitching omitted (except on WW stitching)	117	
	WW stitching: One or more rows of stitching omitted	118	
	Length of stitching less than specified by more than 1/4 inch	119	
Components and assembly	Any component or required operation omitted or not as specified (unless otherwise classified herein)	120	
	Needle chews	121	
	Mends, darns, or patches	122	
	Any turn under on stitched ends of webbing not securely caught in WW stitching	123	
	Turn under on ends of straps, chapes, and loops less than specified by more than 1/8 inch		215
Lowering line pocket	Construction or formation of pocket not as indicated on drawings, obstructing free use of lowering line	124	
Marking and identification	Omitted, incorrect, illegible, misplaced, or size characters not as specified		216
Cleanness	Noticeable spots or stains, i.e., grease, oil, ink, dirt not removed; thread ends not trimmed to 1/4 inch or less		217
Shock absorber	Any cut, tear, or hole	125	
	Cone section not completely adhered (When fabricated separately)	126	
Jacketed wire rope	Not specified type or size	127	
	Ends not nylon coated or wire material not completely closed	128	
	Surface of nylon coated end not smooth or flush with unmelted jacketed surface	129	

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TABLE I. End item visual defects. (cont'd)

Examine	Defect	Classification Major	Minor
Swagging wire rope sleeve	Missing	130	
	Not specified type or size	131	
	Not swagged as specified	132	
	Swagged excessively tight, cutting wire rope material	133	
	Swagged loosely, permitting sleeve to rotate freely	134	

4.4.4 End item dimensional examination. The end items shall be examined for conformance to dimensions on the drawings specified in section 2. Only those dimensions that can be evaluated without damaging or disassembling the end items shall be examined. Any dimension not within the specified tolerance shall be considered a defect. The lot size shall be expressed in units of packs. The sample unit shall be one pack. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

4.4.5 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application
Materials	Any component missing, damaged, or not as specified
Workmanship	Inadequate application of components, such as: incomplete sealing or closure of flap, improper taping, loose strapping, or inadequate stapling
	Bulged or distorted container
Content	Number per container is more or less than required

4.4.6 Palletization examination. The fully packaged and palletized end items shall be examined for the defects listed below. The lot size shall be expressed in units of palletized unit loads. The sample unit shall be one palletized unit load, fully packaged. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

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Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirement
Palletization	Pallet pattern not as specified Load not bonded as specified
Weight	Exceeds maximum load limits
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial, as specified (see 6.2).

5.1.1 Level A preservation. Each jump pack shall be laid flat, felt pad side up. Quick release subassembly, other lines (attached to proper buckle) and two foam end caps shall be laid inside the pack. The pack shall be folded to approximately 38 by 15-1/2 by 7 inches. The pack shall be placed in a plastic bag conforming to A-A-50083 and having a nominal thickness of 0.0040 inch or greater.

5.1.2 Commercial preservation. Each jump pack shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B or Commercial as specified (see 6.2).

5.2.1 Level A packing. Each jump pack preserved as specified in 5.1, shall be packed in a shipping container conforming to overseas type, style A, grade B of PPP-B-601. Inside dimensions of the box shall be approximately 38 inches in length, 15-1/2 inches in width, and 7 inches in depth. Each box shall be closed and strapped in accordance with the appendix of PPP-B-601.

5.2.2 Level B packing. Packs preserved as specified in 5.1, shall be packed in a shipping container conforming to style RSC-L, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. Inside dimensions of the box shall be approximately 38 inches in length, 15-1/2 inches in width, and 7 inches in depth. Each box shall be closed in accordance with method II as specified in the appendix of PPP-B-636.

5.2.2.1 Weather-resistant shipping container. When specified (see 6.2), the shipping container shall be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with the appendix of PPP-B-636.

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top and bottom with container shall have the contents completely covered on the

5.2.3 Commercial packing. Packs, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Palletization. When specified (see 6.2), packs packed as specified in 5.2 shall be palletized on a 4-way entry pallet in accordance with load type I or Ia of MIL-STD-147. Pallet types shall be type I (4-way entry), type IV, or type V in accordance with MIL-STD-147. Pallets shall be fabricated from wood groups I, II, III, or IV of MIL-STD-731. Each prepared load shall be bonded with straps in accordance with bonding means C and D or film bonding means F or G. Pallet pattern shall be number 94 in accordance with the appendix of MIL-STD-147.

5.4 Marking. In addition to any special marking required by the contract or purchase order, unit packs, shipping containers and palletized unit loads shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The jump pack is intended to allow the individual parachutist to carry one Stinger Missile, during a parachute descent.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. When first article is required (see 3.1, 4.3, and 6.3).
d. Levels of preservation and packing (see 5.1 and 5.2).
e. When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).
f. When palletization is required (see 5.3).

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Subject term (keyword listing).

Air delivery
Antiaircraft
Carrier

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Individual equipment
Jump
Weapon case

Custodian:

Army - GL

Review activity:

Army - AV

Preparing activity:

Army - GL

(Project 1670-A777)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-P-44412(GL)	1990 December 11

3. DOCUMENT TITLE
PACK, PARACHUTIST'S, STINGER MISSILE

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
U.S. Army Natick RD&E Center	508-651-4531	256-4531
c. ADDRESS (Include Zip Code) Commander, U.S. Army Natick RD&E Center ATTN: STRNC-UXT Natick, MA 01760-5017	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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